Chapter 11: The Phillips Curve and the Natural Rate of Interest and Output

Gauti B. Eggertsson, Brown University, June 2, 2020

1 Overview

A notable feature missing from the IS-LM model is the lack of any meaningful inflation dynamics, as prices are fixed. Soon after the development of the IS-LM model, economists started documenting that inflation tended to be positively correlated with output. This discovery led to the development of the Phillips curve, named after Phillips (1958), which documented this relationship for Britain. In this chapter we show under what assumptions the model we have considered leads to a Standard Phillips Curve of the form:

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) \tag{1}$$

where $\kappa > 0$ is a coefficient and $\hat{Y}_t^n$ is the *natural level of output*. Relative to the model in the last chapter, where all firms set their price one period in advance, this Phillips curve arises if a fraction γ of firms set their price flexibly, while a fraction $1-\gamma$ index their price to last period's price level. Thus the model nests the case when all prices are flexible ($\gamma = 1$) and the model is "classical" as well as the IS-LM model where all prices are fixed ($\gamma = 0$). As we will see in the next chapter, the standard Phillips Curve came under fundamental criticism by failing to account for the Great Inflation of the 1970s. Nevertheless, it represents the most simple way of introducing price dynamics, which allows us to define several concepts that turn out to be useful in more complex settings such as the canonical New Keynesian model.

With the Phillips curve we introduce some economic concepts that have played a fundamental role in macroeconomic thought for more than a century: the natural rate of interest, the natural level of output and the output gap. I will also show you the closely related efficient rate of interest and the efficient output. Each concept has been defined in a variety of ways over the past 100+ years. The modern definition of the natural rate of interest and the natural level

of output is from Woodford (2003) [4]. While I introduce these concepts in the context of a model that gives rise to (1), they can be applied in most models with price and/or wage rigidities.

The natural rate of interest was first introduced by Knut Wicksell (1898), and it later found its way into the writing of various authors, such as Keynes and Friedman. Currently, it features prominently in the discussion of central bankers. Following Woodford (2003), we define it as the real interest rate if all prices are flexible.

The idea of a natural level of output was first proposed by Friedman (1968) by analogy to Wicksell's natural rate.[1] Again following Woodford (2003), we define it as output if all prices are flexible. The output gap, in turn, corresponds to the difference between equilibrium output and the natural rate of output.

Why are these concepts useful? This natural rate of interest has a useful interpretation that explains its popularity in central banking circles, where it is sometimes referred to as the neutral interest rate or "r-star." I will show that if the central bank sets the nominal interest rate equal to the natural rate, then inflation is fully stabilized. Hence, if the goal of the central bank is to stabilize inflation, the bank should aim to track the natural rate of interest as closely as possible. Moreover, as we will see, this will also stabilize the output gap, defined as the difference between actual output and the natural level of output.

One problem with using the natural level of output as a benchmark is that it is not necessarily the optimal level of output. Another useful concept we will see is the efficient level of output. The efficient level of output measures optimal production, taking into account possible market distortions. This is useful for understanding optimal monetary policy and possible trade-offs between output and inflation, as well as being a useful concept once one considers the jointly optimal monetary and fiscal policy.

2 The Standard Phillips Curve

Consider the same model as in the last chapter, except that we now allow for a time-varying θ_t, for reasons that will become clear in the next section. In this section we streamline the derivation by assuming that $\varphi = \alpha = 0$ (nothing important depends on this and we have a more general specification later on). A fraction γ of firms set their prices flexibly, and their price is denoted as p_t^{flex}.

By solving the problem of the firm following the same steps as in the last

[1] More precisely Friedman introduced *the natural level of employment*. In our model output and employment are related one-to-one by the production function.

chapter, we get the following FOC:

$$\frac{p_t^{flex}}{P_t} = \frac{\theta_t}{\theta_t - 1}\frac{W_t}{P_t}\frac{1}{A_t} \tag{2}$$

Substituting $\frac{W_t}{P_t}$ from the household's optimal labor supply yields[2]:

$$\frac{p_t^{flex}}{P_t} = \frac{\theta_t}{\theta_t - 1}\frac{1}{A_t}(Y_t - G_t)^{\sigma^{-1}} \tag{3}$$

The remaining $1-\gamma$ of firms index their prices to the aggregate price level from last period:

$$p_t^{index} = P_{t-1} \tag{4}$$

Generally the steps you need to follow when deriving the aggregate supply side in a model with price rigidities are to first derive expressions for the price setting of all firms in the economy as in (3) and (4) above, and then substitute the result into the price index, which is given by $P_t \equiv \left[\int_0^1 p_t(i)^{1-\theta_t} di\right]^{\frac{1}{1-\theta_t}}$. Given that there are only two types of price setting by the firms, i.e. p_t^{flex} of mass γ and p_t^{index} of mass $1-\gamma$, the price index is given by:

$$P_t^{1-\theta_t} = \gamma(p_t^{flex})^{1-\theta_t} + (1-\gamma)(p_t^{index})^{1-\theta_t} \tag{5}$$

A first-order approximation of the model is considered in the next proposition:

Proposition 1. ***Standard Phillips Curve.*** *If a fraction γ of firms set their prices flexibly and $1-\gamma$ index to the last-period price index, then the aggregate supply side of the approximated model is summarized by a static Standard Phillips Curve:*

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}_t^n) \tag{6}$$

where $\kappa \equiv \frac{\gamma}{1-\gamma}\sigma_c^{-1}$, $\hat{Y}_t^n = \sigma_c\hat{A}_t + \hat{G}_t - \sigma_c\hat{\mu}_t$.

To obtain equation (6) divide the price index (5) by $P_t^{1-\theta_t}$ and substitute for p_t^{index} and p_t^{flex} using (3) and (4). Linearizing the resulting expression yields the final result. As seen by equation (6), inflation depends upon the difference between output and the natural rate of output, $\hat{Y}_t^n$, which is simply a summary of the shocks in the equation. Before going further, it is useful to define this concept formally, as it has a useful interpretation. An important observation from equation (6), which we will develop in more detail, is that there is no tension between stabilizing inflation and the gap between output and the natural level of output.

[2]Here the assumption of a linear production function in labor and linear disutility of working simplifies our life a little bit

It is useful to observe that the current model nests the IS-LM model we considered as well as the model with flexible prices. If $\gamma \to 0$, then $\kappa \to \infty$, so that $\hat{Y}_t \to \hat{Y}_t^n$ and the model approaches the flexible price limit. Meanwhile if $\gamma \to 1$, then $\kappa \to 0$, so that $\pi_t \to 0$ and prices become fixed, so the model approaches the IS-LM model of last chapter at the other limit. The Old Keynesian Phillips Curve occupies the middle ground.

3 The Natural Level of Output

We now give a more detailed interpretation to the natural level of output. Recall a key result from the last chapter (Proposition 1) was that if prices are flexible, then output is determined independently of the price level and given by equating the household's labor and the firm's labor demand. We restate this condition below, now using the more general notation of last chapter, i.e. allowing for $\varphi > 0$ and $\alpha > 0$:

$$\text{Labor Supply} = \text{Labor Demand}$$

$$\frac{\theta_t - 1}{\theta_t}(1+\tau)(1-\alpha)A_t^{\frac{1}{1-\alpha}}Y_t^{-\frac{\alpha}{1-\alpha}} = \frac{Y_t^{\frac{\varphi}{1-\alpha}}A_t^{-\frac{\varphi}{1-\alpha}}}{(Y_t - G_t)^{-\sigma^{-1}}} \tag{7}$$

Relative to last chapter, we allow for the possibility that the elasticity of substitution θ_t can vary for exogenous reasons. Accordingly there is a new source of exogenous variations (or shocks) in the model. I have also included a new variable $1+\tau$ that can be interpreted as a steady state subsidy for either production or labor supply which is financed via lump-sum taxes on the household. This variable enables us to eliminate steady state inefficiencies stemming from the firms' market power. This is convenient once we define the efficient output. A definition of the natural output follows.

Definition 1. *The natural level of output is the solution $Y_t^n = Y_t$ that solves the model if all prices are flexible, i.e. solves equation (7).*

The firms' markup, $1+\mu_t$, says how much the firms charge for each additional good in excess of the cost of producing it. The markup is then defined as $1+\mu_t \equiv \frac{\theta_t}{\theta_t - 1}$, and we will document variations in θ_t as variations in markups as this has a nicer economic interpretation.

A first-order approximation of (7), using the definition of markups, yields

$$\hat{Y}_t^n = \frac{1+\varphi_\alpha}{\sigma_c^{-1}+\varphi_\alpha}\hat{A}_t + \frac{\sigma_c^{-1}}{\sigma_c^{-1}+\varphi_\alpha}\hat{G}_t - \frac{1}{\sigma_c^{-1}+\varphi_\alpha}\hat{\mu}_t \tag{8}$$

The first important observation about the natural level of output, and this is just a restatement of Proposition 1 from last chapter, is that it is independent

of monetary policy, as envisioned by Friedman (1968). This is not a general result. Consider, for example, the model of last chapter but with money in the utility function that is not additive separable, so that consumption and real money balances are complements (e.g. you get more utility from holding real money balances, the more you consume). In this case, then, output depends on real money balances. As discussed in detail in Woodford (2003) and Galí (2019) [3], however, this specification of money demand complicates the model unnecessarily, and money via this mechanism has trivial quantitative effects in realistic numerical calibrations. The reason is that in modern economies the actual supply of paper currency relative to GDP is very small, and thus incorporating this channel does not have meaningful quantitative effects.

An important feature of the natural level of output is that it changes over time with the exogenous shocks in the model. The exogenous shocks are government spending $\hat{G}_t$, technology (or labor productivity) $\hat{A}_t$, the preference shocks $\hat{\xi}_t$ and variations in markups $\hat{\mu}_t$. It is useful to carefully go through how the shocks change the natural level of output. Notice that the way these shocks affect the natural level of output is essentially the same as in the neoclassical stochastic growth model, or Real Business Cycle model, with two important exceptions. First, the model we are working with abstracts from physical capital. Second, our model incorporates imperfect competition, and thus a new source of fluctuation, via variations in markups. Importantly, this implies that there is a source of output fluctuations that is inefficient.

As equation (8) reveals, the preference shock has no effect on the natural level of output. This is because the preference shock enters multiplicatively in the utility function of the household, and since the labor supply of the household (the right-hand side of (8)) is the ratio of the marginal dis-utility of working to the marginal utility of consumption, it cancels. This is precisely why I have specified the preference shock in this way. It represents a shock that only changes the natural rate of interest, as we will see, without having any effect on the natural output.

If there is a negative productivity shock, $\hat{A}_t$, (think a natural disaster or bad weather, or anything that has an effect on labor productivity), then this will reduce natural output. This is because a negative shock reduces both the demand and supply for labor, i.e. both sides of (8), thus translating into lower output.

An increase in the market power of the firm corresponds to an increase in markups $\hat{\mu}_t$. This reduces the demand for labor by the firms, and hence natural output. You should think of $\hat{\mu}_t$ as a stand-in for various ineffiencies. The Real Business Cycle literature sometimes refers to shocks of this kind as a "labor wedge" because, relative to the standard real business cycle model, it shows

up as a "wedge" in the first-order condition for optimal labor supply on the one hand or in the labor demand of the firm on the other (equation (7)) (see Chari, Kehoe and McGrattan (2007) [2]). Examples include variation in unions' bargaining power and tax distortions that inefficiently reduce firms' incentives to produce or people's incentives to work. A key distinction between variations in $\hat{A}_t$ and $\hat{\mu}_t$ is that the former lead to efficient variations in the natural rate of output, while the latter correspond to inefficient variations. I will clarify this important distinction in the next section.

An increase in government spending increases the natural rate of output. The reason for this is seen in the right-hand side of (7), which corresponds to labor demand. An increase in government spending, for a given output, reduces the consumption available for private consumption. This increases the marginal utility of consumption (the denominator of the right hand equation), leading households to supply more labor. This is the same mechanism as in the RBC literature (see e.g. Baxter and King [1]). Critically, the quantitative importance of this mechanism relies on the elasticity of labor supply, i.e. how much more people are willing to work if their wages are higher. If the household supplies labor inelastically (e.g. if people work 8 hours per day regardless of the wage rate), then government spending has no effect on the natural level of output.

4 The Natural Rate of Interest

As we have seen, the real interest rate, defined as the rate of return on a one-period risk-free real bond denominated in the consumption good, satisfies the consumption Euler Equation:

$$\frac{1}{1+r_t} = \beta E_t \frac{(Y_{t+1} - G_{t+1})^{-\sigma^{-1}}}{(Y_t - G_t)^{-\sigma^{-1}}} \frac{\xi_{t+1}}{\xi_t} \tag{9}$$

The natural rate of interest is then be defined exactly as the real interest rate if prices are flexible:

Definition 2. *The natural rate of interest is the solution $r_t^n = r_t$ that solves the model if all prices are flexible, i.e. it solves equation (9) for $Y_t = Y_t^n$.*

A linear approximation of equation (9) evaluated at $Y_t = Y_t^n$ yields a definition of the natural rate of interest in the approximated equilibrium:

$$\hat{r}_t^n \equiv -\sigma_c^{-1} E_t(\hat{Y}_t^n - \hat{Y}_{t+1}^n) + \sigma_c^{-1} E_t(\hat{G}_t - \hat{G}_{t+1}) + \hat{\xi}_t - E_t\hat{\xi}_{t+1} \tag{10}$$

As this reveals, the natural rate of interest depends on the natural level of output, government spending and the preference shock.

An increase in the natural level of output today, relative to the future, reduces the natural rate of interest. The reason is that higher output today, relative to the future, gives the household an incentive to save, thus reducing the natural interest rate.

Similarly, an increase in government spending, relative to future government spending, increases the natural rate of interest. The reason is that higher government spending takes away resources from private consumption, which leads the household to attempt to increase its borrowing in order to smooth consumption.[3] Since this is a representative household model, no borrowing happens in equilibrium, and hence the interest rate needs to adjust instead to make the households indifferent between current and future consumption. There is a negative effect of higher government spending on the natural rate of interest via the positive effect government spending has on the natural level of output. This effect, however, is always dominated, since the coefficient of spending in equation (8) is less than 1 in absolute value (and this effect disappears if the household supplies labor inelastically).

Finally, the preference shock has a direct effect on the real interest rate as it directly affects the household's utility of consumption today.

4.1 A Useful Gap Representation

Woodford (2003) proposes a useful gap presentation that is generally useful in models with nominal rigidities. The consumption Euler Equation, from the pricing of a one-period risk-free nominal bond, is:

$$\hat{Y}_t - \hat{G}_t = E_t\hat{Y}_{t+1} - E_t\hat{G}_{t+1} - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{\xi}_t + E_t\hat{\xi}_{t+1})$$

If prices are flexible, then equation (9) is:

$$\hat{Y}^n_t - \hat{G}_t = E_t\hat{Y}^n_{t+1} - E_t\hat{G}_{t+1} - \sigma_c(\hat{r}^n_t - \hat{\xi}_t + E_t\hat{\xi}_{t+1}) \tag{11}$$

Subtract the second equation from the first to obtain:

$$\hat{Y}_t - \hat{Y}^n_t = E_t\hat{Y}_{t+1} - E_t\hat{Y}^n_{t+1} - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{r}^n_t) \tag{12}$$

where the natural rate of interest is defined by equation (10).

Recall from last subsection that:

$$\pi_t = \kappa(\hat{Y}_t - \hat{Y}^n_t) \tag{13}$$

[3] Provided the increase in government consumption is temporary. If the increase in spending is permanent, then there is no motive for the household to borrow to increase its consumption. Thus a permanent increase in government spending has no effect on the natural rate of interest via this mechanism.

where $\hat{Y}_t^n$ is now defined more generally in equation (8).

Thus we have summarized the equilibrium in (12) and (13) as the deviation of output from its natural level and interest from its natural level. Defining the output gap as $x_t^n \equiv \hat{Y}_t - \hat{Y}_t^n$ these two equations can be written as:

$$x_t^n = E_t x_{t+1}^n - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{r}_t^n)$$

$$\pi_t = \kappa x_t^n$$

These two equations reveal a fundamental insight that holds more generally: If the central bank wants to target zero inflation, then the nominal interest rate should track the natural rate of interest. Consider the following policy rule:

$$\hat{\imath}_t = \hat{r}_t^n + \phi_\pi \pi_t + \phi_x x_t^n \tag{14}$$

Following is an important proposition:

Proposition 2. *If the central bank tracks the natural rate of interest, as in (14), and* $\phi_\pi + \frac{\phi_x}{\kappa} > 1$*, then a unique bounded solution is* $\pi_t = x_t^n = 0$*.*

What this proposition says is that if the central bank changes the interest rate in lockstep with the natural rate of interest, then it implements full price stability and output fully tracks the natural rate of output.

To prove this proposition, combine the IS and the Phillips Curve and write in terms of the output gap to obtain:

$$x_t^n = \frac{1 + \kappa\sigma_c}{1 + \sigma_c\phi_\pi\kappa + \sigma_c\phi_x} E_t x_{t+1}^n \tag{15}$$

which results in a unique bounded solution solution yielding $\pi_t = x_t^n = 0$ if $\frac{1+\kappa\sigma_c}{1+\sigma_c\phi_\pi\kappa+\sigma_c\phi_x}$ is less than one in absolute value, which is guaranteed by $\phi_\pi + \frac{\phi_x}{\kappa} > 1$. Observe that because the central bank responds to the output gap, it does not require as strong a response to inflation as was mandated with flexible prices where $\phi_\pi > 1$ was required to ensure determinacy. Thus the introduction of endogenous output determination means the central bank can also ensure price stability by reacting to output variations. Observe, moreover, that the values of ϕ_π and ϕ_x do not matter in equilibrium, as long as $\phi_\pi + \frac{\phi_x}{\kappa} > 1$, because inflation is always zero and output is equal to its natural level.

5 The Efficient Output

The natural rate of output is not the level of production that maximizes social social welfare. The efficient output is defined as the output that maximizes the

utility of the representative households, subject to the production possibilities of the economy.

To characterize the efficient output we can solve a *social planner problem.* In constructing a social planner problem, we imagine that the economy is ruled by a benevolent dictator who can tell each agent what to do but is constrained by the production possibilities of the economy (i.e. the resource constraint or production function). Observe that the social planner can operate outside of any price system or market structure, thus he/she does not concern him/herself with prices and wages. The social planner's problem is:

$$\max_{C_t, G_t, N_t} E_0 \sum_{t=0}^{\infty} \{u(C_t) + g(G_t, \delta_t) - v(N_t)\}\xi_t$$

subject to

$$\begin{aligned} C_t &\equiv [\int_0^1 c_t(i)^{\frac{\theta_t - 1}{\theta_t}}]^{\frac{\theta_t}{\theta_t - 1}} \\ G_t &\equiv [\int_0^1 g_t(i)^{\frac{\theta_t - 1}{\theta_t}}]^{\frac{\theta_t}{\theta_t - 1}} \end{aligned}$$

$$c_t(i) + g_t(i) = A_t N_t(i)^{1-\alpha}$$

$$N_t = \int_0^1 N_t(i) di$$

where I have specified general functional forms for utility $u(.)$ and $g(.)$ are increasing and concave in their arguments, while $v(.)$ is increasing and convex. I have now included the utility of government consumption, moreover this utility may vary stochastically via variation in the preference shock δ_t (e.g. due to war or pandemic, etc.). The efficient allocation can be characterized formally by writing an infinite-horizon Lagrangian problem. Since the problem is simple in this setting some shortcuts are available. As all consumption goods are identical, then in the efficient allocation $c_t(i) = C_t$, $g_t(i) = G_t$ and $N_t = N_t(i)$, so that $Y_t = C_t + G_t = A_t {N_t}^{1-\alpha}$. Moreover, the problem boils down to a sequence of static maximization problems, so that we only need to consider the problem in a single period, thus forming the Lagrangian:

$$L_t = u(C_t) + g(G_t, \delta_t) - v(N_t) + \phi_t\{C_t + G_t - A_t {N_t}^{1-\alpha}\}$$

Combining the first-order conditions of this Lagrangian problem yields the following three conditions for the efficient allocation that are sufficient to pin down C_t^e, G_t^e and Y_t^e:

$$u_c(C_t^e) = g_G(G_t^e, \delta_t) \tag{16}$$

$$\frac{v_N((\frac{Y_t^e}{A_t})^{\frac{1}{1-\alpha}})}{u_c(C_t^e)} = (1-\alpha)A_t^{\frac{1}{1-\alpha}} Y_t^{e-\frac{\alpha}{1-\alpha}} \tag{17}$$

$$Y_t^e = G_t^e + C_t^e \tag{18}$$

The first equation says that in the efficient allocation the marginal utility of government consumption is equal to the marginal utility of private consumption. The second says that the marginal disutility of working over the marginal utility of consumption is equal to the marginal product of labor. It is the same relationship as in equation (7) where we equated labor demand and supply, except that in this expression there is no markup or subsidy, as neither has an effect on the production possibilities of the economy. (Recall that the social planner is operating outside of any market or price structure.) These two equations, together with the resource constraint (18), define the efficient level of output.

Definition 3. *The efficient level of output is the output that maximizes social welfare subject to the production possibilities of the economy. It is the value Y_t^e that solves equations (16)-(18).*

The efficient interest rate can be defined in exactly the same way. It is the interest rate that solves (9) for the case in which $Y_t = Y_t^e$.

Substituting for C_t^e using (18) then approximating (16) and (17) yields:

$$\hat{Y}_t^e = \frac{1+\varphi_\alpha}{\sigma_c^{-1}+\varphi_\alpha}\hat{A}_t + \frac{\sigma_c^{-1}}{\sigma_c^{-1}+\varphi_\alpha}\hat{G}_t^e \tag{19}$$

$$\hat{G}_t^e = \frac{\sigma_c^{-1}}{\sigma_c^{-1}+\sigma_g^{-1}}\hat{Y}_t^e + \hat{g}_t^e \tag{20}$$

where $\hat{g}_t^e \equiv -\frac{g_{G\delta}}{\sigma_g^{-1}+\sigma_c^{-1}}\delta_t$ and $\sigma_c \equiv -\frac{u_c}{u_{cc}Y}$, $\sigma_g \equiv -\frac{g_G}{g_{GG}Y}$.

These equations illustrate a critical difference between efficient output and the natural output given by expression (8). The natural rate of output depends on variations in markups while the efficient output does not.

As the efficient equilibrium is derived as a solution to a social planner's problem, it does not specify how it is implemented in a market economy via prices and wage setting. Can it be implemented? We observe by comparing equations (7) and (17) that the efficient and natural output coincide if $(1+\tau) = \frac{\theta}{\theta-1}$, i.e. if there is subsidy on the firms' production or on the household's labor supply to offset the firms' markup.

Equation (20) can be used to solve for the efficient output in terms of productivity and preference shocks only, independent of government spending:

$$\hat{Y}_t^e = \frac{(\sigma_c^{-1}+\sigma_g^{-1})(1+\varphi_\alpha)}{\varphi_\alpha\sigma_c^{-1}+\varphi_\alpha\sigma_g^{-1}+\sigma_g^{-1}\sigma_c^{-1}}\hat{A}_t + \frac{(\sigma_c^{-1}+\sigma_g^{-1})\sigma_c^{-1}}{\varphi_\alpha\sigma_c^{-1}+\varphi_\alpha\sigma_g^{-1}+\sigma_g^{-1}\sigma_c^{-1}}g_t^e$$

The underlying assumption in the efficient allocation is that government spending is set at its optimal level, while no such assumption was made when we derived the natural allocation. The efficient allocation is sometimes more useful to summarize the underlying shocks when one is analyzing both monetary and fiscal policy jointly.

If government spending is always at its efficient level, then the natural and efficient output are related together as follows:

$$\hat{Y}_t^n = \hat{Y}_t^e - \frac{1}{\sigma_c^{-1}+\varphi_\alpha}\hat{\mu}_t \tag{21}$$

i.e. the natural output only differs from the efficient level due to variations in markups.

For welfare evaluation it is useful for technical reasons to consider an economy in which natural output and the efficient output are relatively close. We can approximate the natural rate of output in steady state around the efficient output steady state[4], i.e. approximate the steady state natural output around the point $(1+\tau)=\frac{\theta}{\theta-1}$ using τ as a variable in the expansion using (7). Denoting the steady state efficient output by $\bar{Y}^e$ and steady state natural output by $\bar{Y}^n$ we obtain:

$$\frac{\bar{Y}^e-\bar{Y}^n}{\bar{Y}^e} \approx -\frac{1}{\sigma_c^{-1}+\varphi_\alpha}\hat{\tau} \tag{22}$$

where $\hat{\tau} \equiv \frac{\tau-\bar{\tau}}{1+\bar{\tau}}$. We denote by x^* the gap between efficient output and the natural level in the empirically realistic case in which $\tau=0$, i.e.:

$$x^* \equiv \frac{\theta^{-1}}{\sigma_c^{-1}+\varphi_\alpha} > 0 \tag{23}$$

The variable x^* measures how far steady state natural output is from the steady state efficient output.

We denote by $x_t \equiv \hat{Y}_t - \hat{Y}_t^e$ the output gap expressed in terms of the efficient output. Let us now rewrite the Old Keynesian Phillips curve in terms of the efficient output gap:

$$\pi_t = \kappa x_t + \frac{\gamma}{1-\gamma}\hat{\mu}_t \tag{24}$$

where $\kappa \equiv \frac{\gamma}{1-\gamma}(\varphi_\alpha+\sigma_c^{-1})$.

[4]I am defining a steady state as the constant solution of the model if all shocks are held constant at their average level)

Following the same steps as in the previous section, we can also express the IS curve in terms of the output gap as:

$$x_t = E_t x_{t+1} - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{r}_t^e) \tag{25}$$

where

$$\hat{r}_t^e \equiv -\sigma_c^{-1} E_t(\hat{Y}_t^e - \hat{Y}_{t+1}^e) + \sigma_c^{-1} E_t(\hat{G}_t^e - \hat{G}_{t+1}^e) + \hat{\xi}_t - E_t\hat{\xi}_{t+1}$$

Equations (24) and (25) reveal fundamental insights.

Consider first the solution in the absence of shocks. Both $\hat{Y}_t$ and $\hat{Y}_t^e$ are expressed relative to their own steady state, which was derived under the assumption of no shocks and zero inflation. For $\tau = 0$, the efficient output steady state is higher than the steady state output corresponding to x^*. Thus, if the central bank wants output to be equal to the efficient steady state, it sets $x_t = x^*$, which in the absence of markup shocks implies $\pi_t = \kappa x^*$.

At a fundamental level, this implies that there is a long-run trade-off between inflation and output. The long-run trade-off between inflation and output became a major source of criticism in the 1970s, as we will see in the next chapter. This trade-off, however, also arises in the optimal response of the government to shocks.

Recall that if the central bank targets the natural interest rate, then it fully stabilizes inflation (and the output gap defined in terms of natural output). Equations (24) and (25) suggest, however, that if the central bank tracks the *efficient* interest rate, it does not stabilize inflation. Consider the case in which $\hat{\mu}_t$ is iid mean zero and policy is given by:

$$\hat{\imath}_t = \hat{r}_t^e + \phi_\pi \pi_t + \phi_x x_t \tag{26}$$

and suppose that $\phi_\pi + \frac{\phi_x}{\kappa} > 1$. Substituting the policy rule into the IS equation, and solving using the same steps as in the last subsection, we obtain a unique bounded solution for output and inflation:

$$x_t = -\frac{\gamma}{1-\gamma}\frac{\sigma_c\phi_\pi}{1+\sigma_c\phi_\pi\kappa+\sigma_c\phi_x}\hat{\mu}_t \tag{27}$$

$$\pi_t = \frac{\gamma}{1-\gamma}\frac{1+\sigma_c\phi_x}{1+\sigma_c\phi_\pi\kappa+\sigma_c\phi_x}\hat{\mu}_t \tag{28}$$

which shows that inflation and output both move due to variations in μ_t depending on the relative weight of the two response coefficients ϕ_x and ϕ_π. This shows that there is a trade-off in the short run in response to shocks. If the interest rate response to inflation, ϕ_π, is very large relative to ϕ_x, then markup

variations are only reflected in output, while they only show up in inflation if the response to output, ϕ_x, is high relative to ϕ_π.

But how should the government trade off inflation and output in the long run and in response to shocks? To address this question we need a welfare criterion.

6 The trade-off between inflation and output and optimal policy

What should be the objective of society? We have already derived the efficient level of output, which corresponds to the best society can do in terms of utility. Should policy not just try to approximate this as closely as possible? This was indeed the perspective of some early work on business cycle stabilization, where it was stipulated that the government should aim at achieving "full employment."

That perspective, however, does not take into account the market structure of the economy, or the nature of price setting, which brings in additional complications. A natural social welfare criterion, then, is the utility of the representative household, taking into account the equilibrium behavior of households and firms. For lack of time and space, I will not show here the details of the derivation (see Woodford or Galí's textbooks for a closely related derivation in the Calvo model) but a second-order approximation of the household's welfare in (16) is given by:

$$E_0 \sum_{t=0}^{\infty} \beta^t [\pi_t^2 + \lambda_x (x_t - x^*)^2],$$

where $x^* > 0$ is given by equation (23).

This objective function highlights two distortions which are highly intuitive. The first is associated with inflation. Recall that in the model a fraction of firms set their price flexibly, while a fraction indexed their price to the prior period's price level. Since the firms all share the same production function, and households evaluate each good in the same way, the ideal situation would be that they charge the same price and thus produce the same amount. In case of inflation, however, prices are higher in this period than the last, which means the flexible-price firms are setting a higher price than those that index to past prices. This creates an allocative distortion, as some firms are producing more than others. This allocative distortion is increasing in the level of inflation, which explains the first element. The second part of the objective says that the government would ideally like output to be above its steady state; more precisely it would like equilibrium output to be equal to the efficient output.

The reason for this, as we have already outlined, is the presence of markups. The second term thus captures the monopoly distortion.

With a well-defined social objective in hand, we can now derive optimal policy.

What is an optimal policy? We define it as the policy that maximizes welfare subject to the Phillips curve and the IS equation.

To find it, we formulate the Lagrangian:

$$\mathcal{L}_t = E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \big\{ [\pi_t^2 + \lambda_x (x_t - x^*)^2] + \phi_{1t} (\pi_t - \kappa x_t - \frac{\gamma}{1-\gamma} \mu_t) + \phi_{2t} [x_t - x_{t+1} + \sigma_c (i_t - \pi_{t+1} - r_t^e)] \big\}$$

Remark 1: We can take the expectation operator out of the Lagrangian by using the Law of Iterated Expectations, hence now the expectation appears only outside the Lagrangian. The first-order conditions are:

$$\begin{aligned} \frac{\partial L_t}{\partial \pi_t} &= \pi_t + \phi_{1t} - \beta^{-1} \sigma_c \phi_{2t-1} = 0 \\ \frac{\partial L_t}{\partial x_t} &= \lambda_x (x_t - x^*) - \kappa \phi_{1t} + \phi_{2t} - \beta^{-1} \phi_{2t-1} = 0 \\ \frac{\partial L_t}{\partial i_t} &= \phi_{2t} \sigma_c^{-1} = 0 \end{aligned}$$

Remark 2 The last condition implies that $\phi_{2t} = 0$. What this condition is saying is that the IS equation is not imposing any constraint on the optimal plan: Indeed, we can solve the problem without referring to it (just determining the optimal pair of π_t and x_t). We can simply write the Lagrangian as:

$$L_t = E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \left\{ \left[\pi_t^2 + \lambda_x (x_t - x^*)^2 \right] + \phi_{1t} (\pi_t - \kappa x_t - \frac{\gamma}{1-\gamma} \hat{\mu}_t) \right\}$$

obtaining:

$$\begin{aligned} \frac{\partial L_t}{\partial \pi_t} &= \pi_t + \phi_{1t} = 0 \\ \frac{\partial L_t}{\partial x_t} &= \lambda_x (x_t - x^*) - \kappa \phi_{1t} = 0 \end{aligned}$$

Remark 3: We have already decided to ignore demand for money for exactly the same reason – it is redundant and only relevant to back out the money

supply consistent with the interest rate in the contemplated equilibrium. If we added the money demand to the problem, the Lagrange multiplier on that equation would be zero.

Remark 4: The IS equation is then only useful to back out the interest rate needed to implement the optimal choice of π_t and x_t. Similarly the money demand is only relevant to back out the money supply needed to implement the optimal choice of π_t, x_t and i_t.

Remark 5: Remark 4 is actually a bit "deeper" than it may appear. It is saying that the entire aggregate demand side of the model, i.e. the IS equation, is irrelevant when determining optimal policy in terms of inflation and output! This insight applies more generally, for example in the standard New Keynesian model. The IS equation, however, will start playing a fundamental role once the zero lower bound becomes binding. In that case, the demand side starts playing a central role in the analysis.

Solution: taking account that $\phi_{2t} = 0$, we can combine the FOCs to yield:

$$\frac{\lambda_x}{\kappa}(x_t - x^*) + \pi_t = 0$$

Substitute this into the Phillips Curve to obtain:

$$x_t = \frac{\lambda_x}{\lambda_x + \kappa^2} x^* - \frac{\kappa}{\lambda_x + \kappa^2} \frac{\gamma}{1-\gamma} \hat{\mu}_t$$
$$\pi_t = \frac{\lambda_x \kappa}{\lambda_x + \kappa^2} x^* + \frac{\lambda_x}{\lambda_x + \kappa^2} \frac{\gamma}{1-\gamma} \hat{\mu}_t$$

Hence, the government will tolerate some inflation in steady state, i.e. $\pi_t > 0$, because this allows it to have higher output, i.e. $x_t = \frac{\lambda_x}{\lambda_x + \kappa^2} x^* < x^*$, which is greater than zero, but not quite at the fully efficient rate of output, which would require $x_t = x^*$. Moreover, in response to the shocks μ_t the government will tolerate some inflation in order to achieve better output realization.

Exercise: Consider for example a persistent μ_t shock that follows an AR(1) process. This could have the interpretation, for example, of being a oil shock.

$$\mu_t = \rho \mu_{t-1} + \epsilon_t \tag{29}$$

where $0 < \rho < 1$ and ϵ_t is a iid mean zero shock. What is the dynamic of inflation and the output gap in response to the shock?

References

[1] Marianne Baxter and Bob King. "Fiscal Policy in General Equilibrium". *The American Economic Review* 83.3 (1993), pp. 315–334.

[2] V. V. Chari, Patrick J. Kehoe, and Ellen R. McGrattan. "Business Cycle Accounting". *Econometrica* 75.3 (2007), pp. 781–836.

[3] Jordi Galí. *Monetary Policy, Inflation and the Business Cycle: An Introduction to the New Keynesian Framework and its Applications*. 2nd ed. Princeton University Press, 2015.

[4] Michael Woodford. *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press, 2003.